Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **5 kg** cart starts from rest under **35 N** net force right for **4 s**.
1. **Find acceleration** and final velocity.
2. **State one thing you** are treating as true.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A **3 kg** object in a field of **4 N/kg** has weight **3×4=12 N**. Moving it to a **2 N/kg** field halves weight to **6 N** but leaves mass **3 kg**.

Your turn: Draw weight arrows for the two places and **label the unchanged quantity**.

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The same **2 kg** object visits worlds with g=**10** and g=**5 N/kg**. **Compare its mass** and weight.

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

On World B, g = **5 N/kg**. **Complete the equipment table**. **Explain why doubling the packed** mass doubles its weight.

Equipment mass (kg)	Weight (N)
3	____
6	____

Marzano check - circle: ① ② ③

Weight and local gravity

antocciphysics

05 Explain your reasoning

A scale reports **0 N** for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force? Explain.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** object has **32 N** right and **8 N** left. **Draw the forces** and **find net force** and acceleration.
Draw the forces on the chosen object and **label their sources**.

Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

A **4 kg** object is where g=**3 N/kg**. **State its mass** and weight.

Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Two objects weigh the same on different worlds. Must their masses be the same? Build a numerical counterexample.

Case	Mass (kg)	g (N/kg)	Weight (N)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③